Shi Yang Ng

Mobile No: +65 89041430 | Email: ngsh0069@e.ntu.edu.sg
Github: https://github.com/shiyangng | LinkedIn: https://www.linkedin.com/in/shi-yang-ng-0237242ba/
Portfolio: https://shiyangng.github.io/personal-portfolio/

EDUCATION

Nanyang Technological University, Singapore **Aug 2024 - May 2028**
Bachelor of Electrical and Electronic Engineering with a Second Major in Data Analytics
Dean's List AY24/25
Honours (Highest Distinction), Current CGPA: 4.72/5.00
Relevant Modules: Data Structures & Algorithms in Python, Analytics I: Visual & Predictive Techniques, Introduction to Data Science & Artificial Intelligence, From Computational Thinking to Programming

WORK EXPERIENCE

Nexperia, Guangdong, China **May 2025 - Aug 2025**
Software Developer Intern

- Developed a full-stack application with a clean dashboard UI using Ant Design, React and TypeScript, reduced file recovery time and improved system reliability through an intuitive version control interface.
- Optimised backend performance by integrating REST APIs and SMB Protocol with NestJS and PostgreSQL, enabling seamless cross-platform file sharing for 250+ concurrent users while reducing data access latency.
- Engineered an automated data pipeline to clean and standardise system metadata, leveraging Python and SQL to build a real-time performance dashboard that provided leadership with actionable insights into storage efficiency.
- Initiated and delivered a comprehensive technical presentation using Canva on Generative AI technologies and automation workflows for 20+ employees to drive internal AI awareness.

PROJECTS

PathForger - AI Workforce Growth Platform (PSA Hackathon), Lead Developer **Oct 2025 - Oct 2025**

- Built an AI-driven system using FastAPI and modular LLM agents for personalised career recommendations and mentorship matching in a 72-hour hackathon.
- Designed vector intelligence pipelines and real-time user friendly dashboards using React and TypeScript to analyse workforce skills, readiness, and engagement.
- This AI-powered platform provides personalised career development, intelligent mental support, predicts leadership potential,which enables inclusive growth

SimplifyNext x AWS Agentic AI Hackathon 2025, Team Leader **Sep 2025 - Sep 2025**
SuperConfig Agent

- Engineered a no-code configuration system using AWS bedrock to deploy personalised agentic AIs in under 5 minutes, supporting 5+ enterprise use cases (ticketing, scheduling, knowledge retrieval, daily digests, web search).
- Implemented an AI backend utilising RAG, Strands, AWS Bedrock, vector databases, SQL, multi-agent orchestration, Docling, MCP, and FastAPI, with a React + Tailwind frontend and Telegram integration
- Delivered key AI capabilities including ticketing automation, intelligent meeting scheduling, company knowledge retrieval, daily digest generation, and real-time web search.
- Reached to semifinals of SimplifyNext x AWS Hackathon, with finals preparation ongoing.

Nanyang Technological University, Singapore **Mar 2025 - May 2025**
Data Analytics Mini Project, Team Leader

- Developed a Python-based machine learning model to identify hidden high-risk cardiovascular subgroups among individuals with normal BP.
- Executed end-to-end data preprocessing, exploratory data analysis (EDA), and forward feature selection.
- Strategised model selection by comparing Random Forest, Logistic Regression, and SVM models to extract actionable insights for early CVD risk detection.

Portfolio Website, Self Initiated

- Designed and created a responsive personal website deploying ReactJS (JSX), TailwindCSS, and Framer Motion, featuring striking animations and a clean UI.
- Showcases academic and AI/ML projects, revamping professional visibility and serving as a centralised platform for recruiters and collaborators.

EcoEats, Self Initiated

- Constructed a full-stack web application with React, TailwindCSS, Ant Design and Firebase for a smooth user experience, promoting sustainable eating by sharing eco-friendly recipes and lifestyle tips.
- Implemented user–admin authentication, order management, and payment integration with Stripe API to boost platform functionality and usability.

WeatherApp, Self Initiated

- Built a responsive web app displaying real-time weather conditions for any city using the AccuWeather API, with Firebase for deployment and data handling.
- Designed a clean and intuitive UI with HTML, CSS, and JavaScript, delivering instant city search and sub-second API response times for a smooth user experience.

CO-CURRICULAR & VOLUNTARY ACTIVITIES

NTU **Aug 2025 - Present**
Hall Chess Captain

- Strategised team preparation and match tactics to secure a Top 5 placement out of 23 halls during InterHall Recreational Games
- Engaged junior players through weekly mentoring and training sessions, fostering a competitive and united team spirit

SKILLS & INTERESTS

Technicals: React, NestJS, JavaScript, TypeScript, Python, R, C, PHP, SQL, PostgreSQL, NodeJS, Firebase, Bash, HTML, CSS, ExpressJS, REST APIs
AI & ML: AWS Bedrock, RAG, Hugging Face, TensorFlow, LangChain, LangGraph, Vector Databases, Strands, MCP, mem0, Pandas, NumPy, Matplotlib, ggplot2, Plotly, Seaborn
Design: Figma, Canva, PowerPoint, TailwindCSS, Ant Design, Bootstrap
Languages: Proficient in English, Chinese, Malay